

TEAM CYBERPULSE

ACCESSICITY

Wheelchair & Stroller Routing

Crowd-mapping urban infrastructure to deliver step-free navigation for vulnerable citizens.

OFFICIAL PROJECT MOTTO

"DON'T JUST FIND A ROUTE. FIND A ROUTE YOU CAN ACTUALLY USE."

HANDBOOK PURPOSE

This document is an exhaustive, student-focused technical revision booklet and judge cross-examination handbook. It breaks down the core architecture, graph-routing math, OpenStreetMap integration, crowdsourced verification model, computer vision support layer, and answers **50 critical judge attack scenarios** for Ideathon 2026.

1. What Exactly is AccessiCity?

FOUNDATION

Simple English: AccessiCity is an accessibility-aware navigation platform that leverages community contributions and smart filtering to calculate step-free, usable routes for wheelchair users, parents with strollers, and citizens with mobility constraints.

Hinglish Mein: Hum map ko sirf distance ya traffic ke liye nahi, balki physical accessibility ke liye use karte hain. Normal map direction dikhatay hain, AccessiCity unusable barriers ko filter karta hai.

Mental Model: Normal Maps = "Where is the destination?" | AccessiCity = "Can I physically travel this path?"

REMEMBER THIS: It is an accessibility layer on top of navigation, not just a static directory of ramps.

2. What is the Core Problem?

PROBLEM STATEMENT

Simple English: Urban navigation fails vulnerable groups because single physical barriers—such as a 5-step staircase, broken metro elevator, steep curb, or blocked ramp—render an entire calculated route physically impossible to navigate.

Hinglish Mein: Map par route technically present hai, lekin ek wheelchair user ke liye woh route practically exist hi nahi karta. Single sidhi (stair) poora rasta block kar deti hai.

Judge Note: Judges will ask if this is a real issue. Answer: "Yes, routing without accessibility features creates physical dead-ends."

3. Who Faces the Problem? (Target Audience)

USER BASE

PRIMARY USERS (Core Focus)

- Wheelchair users (manual & motorized)
- Parents with baby strollers / prams

SECONDARY USERS (Future Scale)

- Elderly citizens with reduced mobility
- People with temporary injuries (crutches/casts)

REMEMBER THIS: Always distinguish primary MVP users from broader demographic expansions during pitches.

4. Why are Normal Navigation Apps Not Enough?

GAP ANALYSIS

Standard engines optimize for **Distance (meters)** and **Time (seconds)** over paved roads. They lack granular micro-pedestrian data like staircase step counts, elevator operational state, curb cuts, surface roughness, and ramp incline gradients.

Hinglish Mein: Google Maps yeh toh bata dega ki Metro Station 200m door hai, lekin yeh nahi batayega ki Gate No. 2 ki lift kharab hai aur Gate No. 3 par 12 sidhiyaan hain.

5. Difference From Google Maps

KEY DIFFERENTIATOR

"Google Maps tells you where to go. AccessiCity tells you whether you can actually get there."

Google Maps treats pedestrian walkways as uniform lines. AccessiCity introduces *accessibility as a hard constraint* in the route calculation matrix.

6. What Does "100% Step-Free Path" Mean?

CORE CONCEPT

A route mathematically filtered to contain **zero non-ramped steps or non-negotiable physical vertical barriers** based on verified map nodes.

Judge Defense Strategy:

Do NOT claim 100% absolute physical magic in the real world. Say: *"We guarantee a 100% step-free path according to our verified data pipeline and real-time community reports."*

REMEMBER THIS: Always frame 100% step-free as a constraint filter applied over available data.

7. How Does the Complete Workflow Work?

SYSTEM WORKFLOW

1. User Inputs
Destination

2. Select Step-Free
Profile

3. Filter Ramp/Lift
Nodes

4. Live Elevator
Check

5. Accessible
Route Generated

The engine retrieves map graph, strips edges containing invalid step tags or reported broken elevators, evaluates computer-vision verified ramp nodes, and returns the optimized graph path.

8. What Happens When Destination is Entered?

EXECUTION FLOW

1. **Client App:** Captures origin GPS, destination coordinate, and mobility profile.
2. **API Request:** Sends payload to Backend Routing REST Gateway.
3. **Backend Engine:** Queries Spatial DB, pulls OpenStreetMap nodes + crowdsourced layer.
4. **Graph Filtering:** Removes bad nodes; executes Dijkstra / A* with penalty weights.
5. **Client Render:** Returns GeoJSON polyline with turn-by-turn accessibility callouts.

9. What is the Accessibility Layer?

DATA ARCHITECTURE

It is a dynamic geospatial data layer overlaying standard vector map base tiles.

Hinglish Mein: Jaise Google Maps ke upar Traffic ka red/green layer hota hai, waisa hi humara Accessibility Layer hai jo ramps, lifts, aur stairs ki condition dikhata hai.

[BASE MAP TILES (OSM)] + [ACCESSIBILITY OVERLAY (Ramps/Lifts/Stairs)] = ACCESSI-ROUTABLE GRAPH

10. OpenStreetMap (OSM) & Why Use It?

MAP FOUNDATION

OpenStreetMap is an open-source, editable geographic database of the world. It provides raw spatial geometry (nodes, ways, relations) allowing us to inject custom accessibility attributes without paying expensive commercial API lock-in fees.

11. What are Accessibility Tags?

DATA FORMAT

Key-value metadata pairs attached to geospatial elements in the OpenStreetMap schema.

OSM Key	Value	AccessiCity Meaning
wheelchair	yes / no / limited	Is feature wheelchair accessible?
highway	steps / footway	Identifies physical stairs vs smooth path
incline	% or degree	Gradient steepness (Crucial for manual wheelchairs)

12. What is a Node and What is an Edge?

GRAPH THEORY

NODE (Point)

A specific geographic point in space. e.g., Entrance, Elevator door, Ramp start, Stairwell base, Junction.

EDGE (Connection)

The path segment connecting two nodes. Has attributes like distance, slope, and surface type.

Hinglish Mein: Nodes matlab destinations/junctions (points), aur Edges matlab unhe jodne wale raste (lines).

13. How Does Routing Work Technically?

ALGORITHM

The navigation space is converted into a weighted graph $G = (V, E)$. Standard algorithms (Dijkstra / A*) calculate shortest path. AccessiCity dynamic routing assigns an **infinite cost weight (∞)** or removes edges that violate accessibility rules.

$$\text{Path Cost} = \text{Distance} + \text{Penalty}(\text{Stairs}) + \text{Penalty}(\text{Incline}) + \text{Penalty}(\text{Unverified})$$

14. What is a Routing Constraint?

LOGIC RULE

A logical rule or condition that a path segment *must* fulfill to be included in the final route computation. If `Profile = StepFree` and `Edge.contains(Stairs) == True`, Edge is excluded.

15. How are Stairs Handled?

BARRIER FILTER

When an edge is tagged as `highway=steps` without an adjacent verified ramp or elevator node, its traversal weight is set to infinity for step-free routing profiles, forcing the engine to evaluate alternate longer loops.

REMEMBER THIS: Excluding stairs is hard constraint filtering, not just a preference warning.

16. How are Ramps Handled?

ACCESSIBLE NODE

Ramps are stored as bypass nodes connecting different elevation edges. A verified ramp node cancels out the step penalty of a vertical level change.

Verified Ramp Nodes ensure steep staircases are mathematically replaced by adjacent accessible inclines.

17. How are Elevators Handled?

VERTICAL TRANSIT

Elevators are represented as vertical connection edges between distinct height layers (e.g., Street level to Metro Underground platform). They require an active operational flag to remain usable in routing.

18. What Does "Live Elevator Status" Really Mean?

CRITICAL CONCEPT

It means maintaining a temporal status score based on **recent crowdsourced reports, timestamps, and potential partner IoT/institutional feeds.**

Judge Defense: "Do you really have live elevator sensors in every building?"

Answer: "No, 'Live' refers to fresh temporal crowdsourced telemetry backed by timestamps. If an elevator was reported operational 10 mins ago by 3 users, confidence is high. If unreported for 3 days, status degrades to unverified."

19. How Does Crowdsourcing Work?

COMMUNITY DATA

Citizens upload photos, flag broken elevators, report temporary construction obstacles, or confirm existing ramp conditions using a quick tap interface in the app.

Hinglish Mein: Jaise Waze app par log police ya traffic jam report karte hain, waisay hi AccessiCity par log ramps aur kharab lifts report karte hain.

20. How Do You Prevent Fake or Wrong Reports?

JUDGE FAVORITE

- **Multi-User Consensus:** A single report flags a status as "Pending"; requires N consensus votes to alter official status.
- **Contributor Reputation:** High-reputation users have higher weighted votes.
- **EXIF & Geofencing:** Photos must match user GPS location and time header.

REMEMBER THIS: Data freshness & reliability are managed via consensus algorithms, not blind trust.

21. What is Computer Vision Doing?

AI PIPELINE

A lightweight object-detection model (e.g., YOLO / MobileNet) running on uploaded community photos to automatically tag candidate ramps, detect steps, and flag steep inclines.

Flow: User photo → CV Model scans image → Detects Ramp (Confidence 92%) → Pre-fills tag for verification.

22. Why Shouldn't AI Be the Final Authority?

DESIGN PHILOSOPHY

A photo can show a perfectly constructed ramp, but cannot reveal if the door at the top is locked, or if the surface is dangerously slippery.

GOLDEN DEFENSE LINE: "AI assists the data pipeline; it does not become the final authority. Verification lies in community consensus."

23. Community Contributor Point System

GAMIFICATION

Gamification incentives (badges, civic leaderboards, impact metrics) reward users for verifying nodes and reporting obstacles, driving crowdsourced data longevity.

24. How Does the Frontend Work?

CLIENT LAYER

Built using modern cross-platform frameworks (React Native / Flutter) or web mapping suites (Mapbox GL / Leaflet). Renders vector tiles, handles camera interaction, displays step-free polyline overlays, and presents crowdsourcing camera upload sheets.

25. How Does the Backend Work?

SERVER LOGIC

Powered by Node.js / Python FastAPI server. It acts as the orchestration layer: receives routing requests, queries spatial database, runs graph routing algorithms with accessibility filters, and manages user reputation DB.

Hinglish Mein: Frontend asks (Sawaal karta hai). Backend processes (Dimaag lagata hai aur routing calculate karta hai).

26. How Do Frontend and Backend Communicate? (API)

API GATEWAY

Via RESTful JSON endpoints or WebSockets for live status updates.

```
REQUEST: POST /api/v1/route { origin: [28.53, 77.39], dest: [28.54, 77.40], profile: "step_free"
}
RESPONSE: 200 OK { distance_m: 450, steps_count: 0, verified_ramps: 3, waypoints: [...] }
```

27. What Does the Database Store?

SPATIAL DB

A Spatial Database (PostgreSQL with PostGIS plugin):

- Spatial Node Coordinates (Ramps, Lifts, Obstacles)
- Verification Status, Timestamps, Report Logs
- User Accounts, Reputation Score, Contributions

28. What is the MVP and Why is it Feasible?

FEASIBILITY

MVP Scope: A single high-density urban transit corridor (e.g., Galgotias University Campus OR 2 major Delhi Metro interchange stations).

DEFENSE LINE: "We don't need to map an entire city on day one. We establish proof-of-concept in a closed hyper-frequent node."

29. What is Actually Innovative & Impactful?

INNOVATION

Innovation isn't just "putting ramps on a map"—it is **dynamically binding crowdsourced spatial accessibility tags into graph route computation algorithms** in real-time.

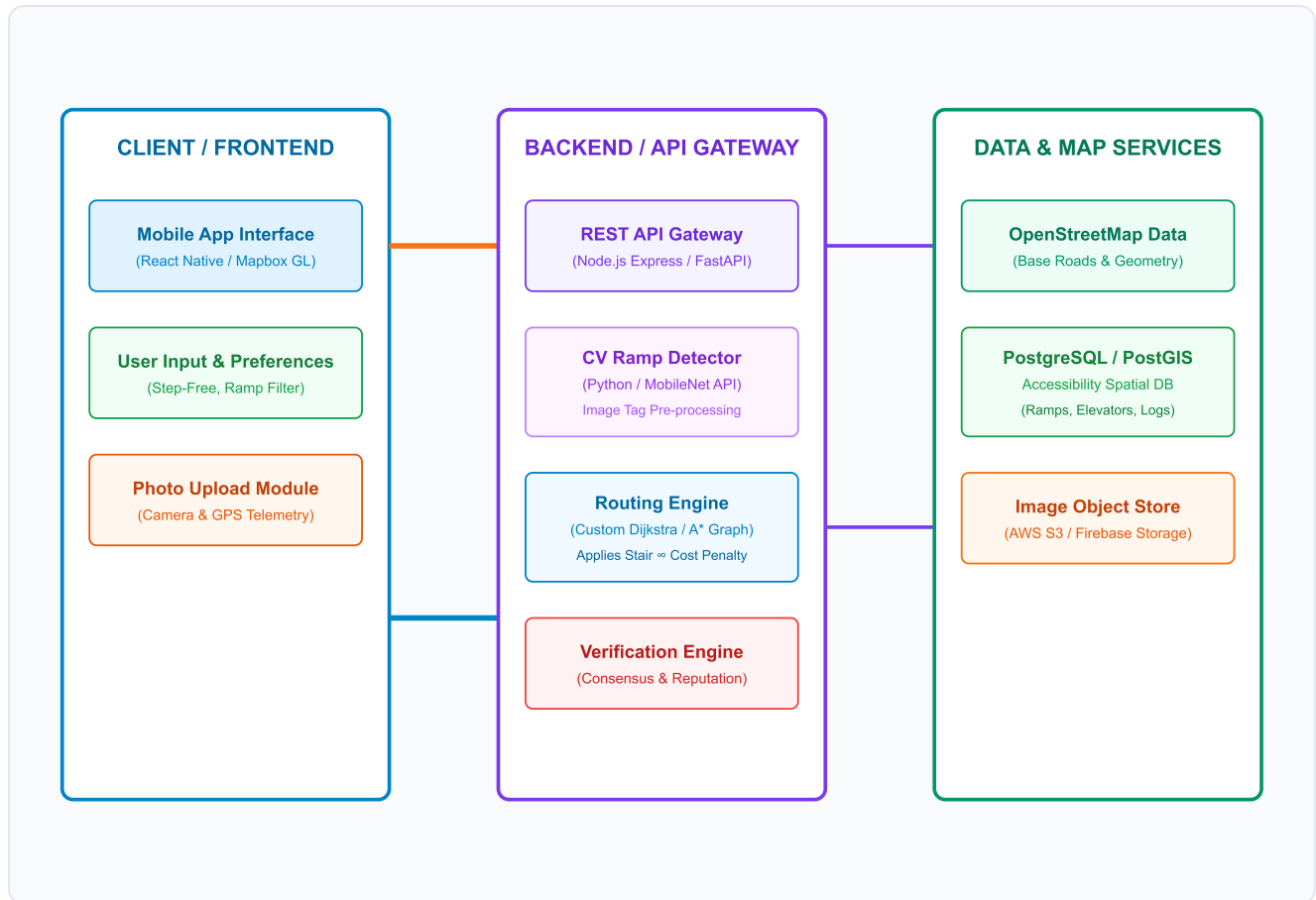
30. What Should I Say If Judges Attack the Idea?

ATTACK SURVIVAL

Acknowledge real-world data limitations gracefully, explain the consensus model, focus on the focused MVP scope, and finish with conviction:

"Don't just find a route. Find a route you can actually use."

The diagram below illustrates the complete data flow from mobile client interaction down to graph route generation and spatial database query pipelines.



Architecture Core Flow Explanation:

When the user requests a route, the **Backend API Gateway** queries base map edges from **OpenStreetMap** and merges them with spatial accessibility nodes from **PostGIS**. The **Routing Engine** strips invalid stair edges and evaluates reported elevator status before pushing the calculated step-free polyline response back to the client app.

SEC 5

FRONTEND SCREEN FLOW & INTERFACE DESIGN

MOBILE UI MOCKUP

📍

Origin: Rajiv Chowk Gate 2

🎯

Destination: Platform 3

Filter: 100% Step-Free

ACTIVE

Accessible Route Found

Distance: 280m • Est. Time: 4 mins

✓ Ramp Node #102 Verified

✓ Metro Lift B Status: OPERATIONAL (12m ago)

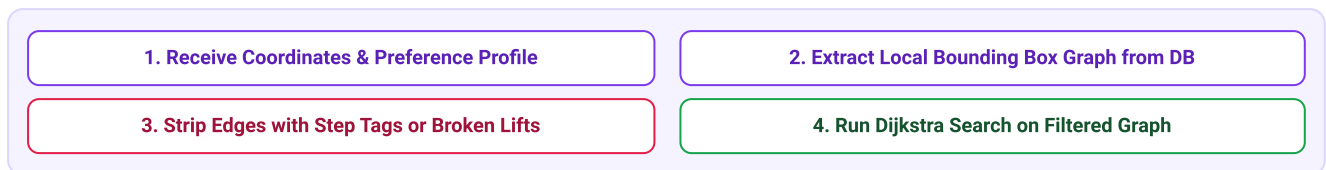
✗ Stairs Bypass Activated

Key UI Components:

- **Step-Free Toggle:** Hard switch triggering specific constraint profile parameters.
- **Ramp & Lift Markers:** Color-coded visual map pins showing node operational status (Green = Verified, Amber = Unverified, Red = Out of order).
- **Quick Crowd-Report Button:** Floating CTA allowing users to instantly upload a barrier photo in 2 taps.

SEC 6

BACKEND ROUTING LOGIC PIPELINE



Technical Term	Simple English Definition	AccessiCity Project Example
API (Application Programming Interface)	Software messenger transferring requests and data between app screen and backend servers.	Frontend requests route payload; Backend API returns JSON polyline with verified ramps.
Node	A discrete spatial coordinate point in a map network representing a location feature.	Elevator entrance node, street junction node, or ramp start node.
Edge	A connecting segment or pathway line between two map nodes.	The footway sidewalk connecting Building A door to Gate B ramp.
Graph	A mathematical network structural representation made of Nodes connected by Edges.	The entire pedestrian map network converted into $G = (V, E)$ for route calculation.
Routing Constraint	A conditional rule filtering out invalid path segments during route calculation.	Setting step edge traversal weight to Infinity when user selects Step-Free profile.
OpenStreetMap (OSM)	An open-source, editable world spatial mapping dataset foundation.	Used as base map framework to extract road geometry and attach custom accessibility tags.
Computer Vision (CV)	AI algorithms processing visual pixels from images to identify real-world objects.	MobileNet model scanning user uploaded photos to auto-detect potential wheelchair ramps.
Data Freshness	A measure of how recently spatial data or operational status was validated.	Elevator status reported 15 minutes ago has higher freshness score than 3-day old data.
Consensus Model	A multi-user voting mechanism required to validate crowdsourced status reports.	Requiring 3 independent user photo confirmations before marking an elevator as broken.
Minimum Viable Product (MVP)	The simplest functional version of a product used to test and validate core assumptions.	Deploying AccessiCity solely within Galgotias University campus or 2 Metro Interchange hubs.

Hinglish Summary: Technical terms se darne ki zaroorat nahi hai. Node = Point, Edge = Line, Constraint = Rule, CV = Photo Reader, Consensus = Crowd Verification.

Q1: Isn't this idea just Google Maps with ramps?

Best Answer: Google Maps prioritizes vehicular and standard pedestrian speed over granular micro-accessibility. AccessiCity builds accessibility directly into the graph engine as a hard routing constraint, discarding paths that contain unmapped physical barriers.

Why works: Acknowledges Google's strength while establishing our specialized domain focus.

Q2: Why can't Google Maps simply implement this feature tomorrow?

Best Answer: Large mapping providers rely on automated satellite imagery and vehicular road data, which cannot capture micro-pedestrian details like elevator breakdown status or step counts inside stations. Our crowdsourced, hyper-local verification model fills this precise data void.

Why works: Highlights structural limitations of satellite-based mapping engines.

Q3: Where do you get your baseline map data from?

Best Answer: We source base geospatial geometry from OpenStreetMap (OSM) and layer our custom PostGIS accessibility database on top using community contributions.

Why works: Shows open-source technical feasibility without reinventing map generation.

Q4: What if your crowdsourced data is wrong or outdated?

Best Answer: We utilize temporal decay scoring. Unverified status confidence degrades over time, prompting community re-verification requests to maintain high data freshness.

Why works: Proves we have thought about data lifecycle management.

Q5: How do you prevent users from submitting fake obstacle reports?

Best Answer: Single reports flag nodes as "Pending". Status changes require multi-user consensus, geotag EXIF validation, and contributor reputation thresholds.

Why works: Demonstrates robust system security logic.

Q6: What happens if an elevator breaks down right after a user starts navigating?

Best Answer: If a real-time community alert is logged, the active app session triggers a dynamic route recalculation prompt bypassing that elevator node.

Why works: Proves real-time event-driven app reactivity.

Q7: Is your elevator status truly "Live"?

Best Answer: It is community-live based on recent timestamped telemetry and available partner APIs (e.g., Metro transit authority updates).

Why works: Honest distinction between hardware sensors and crowdsourced telemetry.

Q8: How accurate is your Computer Vision Ramp Detector?

Best Answer: CV acts as a data intake filter with ~85-90% classification accuracy, pre-filling submission tags that are then validated by community consensus.

Why works: Avoids fake 100% AI perfection claims.

Q9: What if a ramp is too steep for a manual wheelchair?

Q11: Can this platform work effectively in Indian cities?

Best Answer: Yes, particularly in high-density organized transit zones like Delhi Metro interchanges, modern railway stations, malls, and university campuses where accessibility infrastructure exists but status awareness is zero.

Q12: How do you plan to scale mapping to an entire city?

Best Answer: Through a hub-and-spoke expansion strategy—starting with enclosed public hubs (Metro stations, campuses) before expanding out to surrounding street sidewalks.

Q13: Why would a normal non-disabled citizen contribute photos?

Best Answer: Through gamification, civic rewards, partnerships with university volunteer programs (NSS), and parents advocating for stroller accessibility.

Q14: What is your exact Minimum Viable Product (MVP)?

Best Answer: A focused pilot mapping Galgotias University campus and 2 major metro interchange hubs with step-free graph navigation and photo reporting.

Q15: What is the biggest technical challenge in this architecture?

Best Answer: Data sparsity and maintaining high freshness confidence scores across rapidly changing urban environments.

Q16: Which database stack are you employing and why?

Best Answer: PostgreSQL with PostGIS extension for optimized geospatial spatial indexing (R-Tree / GiST indexes) and spatial queries.

Q17: What API protocol connects frontend and backend?

Best Answer: RESTful APIs for route payload exchanges and WebSocket connections for push notifications on live barrier updates.

Q18: Explain how a graph represents a multi-story metro station.

Best Answer: Floor levels are horizontal sub-graphs linked vertically by elevator edge segments or stair edge segments.

Q19: How does the algorithm penalize stairs technically?

Best Answer: By assigning an edge weight multiplier $w_{\text{stairs}} = \infty$ when the user profile requires step-free routing.

Q20: Why rely on OpenStreetMap instead of Google Maps API?

Best Answer: OSM provides raw spatial vector data that we can directly modify, host, and annotate with custom accessibility tags without API rate charges.

Q21: How will you monetize this project in the future?

Best Answer: B2B / B2G API licensing to transit authorities (e.g., Metro rail apps), corporate campus mobility integration, and municipal accessibility auditing analytics.

Q22: Who are your primary target enterprise clients?

Best Answer: Urban transit corporations, large multi-building university systems, international airports, and commercial mall operators.

Q23: What metrics will prove your MVP was successful?

Best Answer: Route completion success rates, accuracy percentage of reported barrier nodes, and community photo contribution retention rates.

Q24: What happens if a user reports a temporary construction block?

Best Answer: The affected spatial edge is tagged with a temporary penalty flag and timestamp expiration limit (e.g., 24-hour decay).

Q25: Why focus on strollers in addition to wheelchairs?

Best Answer: Stroller users represent a massive everyday demographic with identical physical step-free routing constraints, significantly widening app adoption.

Q26: What is the broader social impact of AccessiCity?

Best Answer: Restoring independent urban mobility to vulnerable citizens and highlighting physical infrastructure gaps for municipal planners.

Q27: What happens if your system gives a wrong route and a user gets stuck?

Best Answer: App provides an immediate "Report Barrier / Request Assist" button that logs exact coordinates to rapidly update the graph engine for all subsequent users.

Q28: What are the main limitations of your proposed solution?

Best Answer: Initial dependence on community density for data fresh updates and reliance on public infrastructure existing in some form.

Q29: What would you build first if given 30 days and developer resources?

Best Answer: The PostGIS accessibility schema, custom A* routing constraint filter, and a basic camera upload app for Galgotias campus.

Q30: What is the single biggest risk to this project?

Best Answer: Cold-start data sparsity in new unmapped geographical regions before community network effects activate.

Q31: Pitch your entire project clearly in 30 seconds.

Best Answer: "AccessiCity is a crowd-mapped navigation platform for wheelchair and stroller users. We collect verified micro-accessibility data—like ramps and working elevators—and plug it directly into graph routing algorithms to deliver 100% step-free routes. Don't just find a route. Find a route you can actually use."

Q32: Is Computer Vision strictly necessary or just a buzzword?

Best Answer: It reduces manual user reporting friction by 70% by automatically classifying features from photos, significantly improving data ingestion speed.

Q33: How do you handle multi-floor elevation maps inside complex buildings?

Best Answer: By building a 3D-aware graph where nodes possess an (x, y, z) level attribute connected vertically by elevator edges.

Q34: How does your app help municipal governments?

Best Answer: Generates heatmaps of urban accessibility bottlenecks, showing urban planners exactly where ramps or elevator repairs are urgently needed.

Q35: Can visually impaired users benefit from this platform?

Best Answer: Secondary expansion profiles can filter for tactile paving paths and audio-assisted crossing signals.

Q36: Why not just rely on government accessibility audits?

Best Answer: Government audits are static and performed once every few years, whereas infrastructure state changes daily (e.g., lift breakdowns).

Q37: What security measures protect user location privacy?

Best Answer: Trip origin and destination endpoints are anonymized, and crowdsourced photo submissions strip EXIF personal identifiers while retaining spatial coordinates.

Q38: How does the system handle temporary ramp obstructions like parked cars?

Best Answer: Users flag a temporary barrier; system applies a dynamic edge penalty that auto-expires after 4 hours unless re-confirmed.

Q39: What makes your team (CYBERPULSE) qualified to build this?

Best Answer: Combining CSE Core software expertise with spatial database understanding, algorithm logic, and personal passion for inclusive urban design.

Q40: How does AccessiCity integrate with smart city initiatives?

Best Answer: Serving as an open accessibility data API feed for municipal transit apps and public smart city dashboards.

Q41: What if a user has an electric wheelchair vs manual wheelchair?

Best Answer: Profiles allow parameter customization: electric wheelchairs can handle slightly steeper inclines, whereas manual profiles enforce strict max-gradient limits.

Q42: What algorithm optimization ensures sub-second route calculation?

Best Answer: Spatial graph bounding and Contraction Hierarchies (CH) pre-processing to compute path queries in milliseconds.

Q43: How do you verify photo uploads without manual review?

Best Answer: CV object confidence thresholds combined with automated cross-verification from localized user consensus.

Q44: What is your contingency if OpenStreetMap data is missing in an area?

Best Answer: App enables active crowd-mapping mode, allowing early adopters to record path vectors and trace accessible corridors.

Q45: How will you acquire your first 1,000 active contributors?

Best Answer: Partnering with university volunteer groups, disability advocacy NGOs, and Metro commuter communities.

Q46: Is this solution patentable or defensible?

Best Answer: Defensibility lies in network effects, real-time proprietary crowdsourced spatial data assets, and localized verification algorithms.

Q47: Why focus on urban metro interchanges first?

Best Answer: They represent high-density multi-level bottlenecks where elevator availability directly dictates commuter transit success.

Q48: How does weather affect accessibility data reliability?

Best Answer: Rain/flooding flags can temporarily mark outdoor unpaved footpaths or steep uncovered ramps as hazardous/impassable.

Q49: What is the total addressable market for accessibility tech in urban zones?

Best Answer: Over 15% of the global population lives with mobility limitations, coupled with millions of families using strollers daily.

Q50: What is the core takeaways you want judges to remember today?

Best Answer: "AccessiCity makes accessibility a first-class routing constraint. We transform open spatial data into usable, barrier-free urban mobility."

SECTION 9: IF THE JUDGE SAYS "THIS IS JUST GOOGLE MAPS"

Scripted Response: "Sir, Google Maps already solves macro navigation extremely well. Our proposal is not trying to replace Google Maps. Our focus is a specific, acute problem where physical accessibility becomes a hard routing constraint. We combine micro-spatial accessibility tags, crowdsourced status verification, and constraint-based graph search so the calculated route is selected based on whether a wheelchair user can physically use it."

Why Strong: Respects existing platforms while highlighting our hyper-specialized algorithm constraint layer.

SECTION 10: IF THE JUDGE SAYS "YOUR AI / CV IS FAKE OR UNRELIABLE"

Scripted Response: "We are intentionally NOT depending entirely on AI. Computer vision is strictly an assistance layer to reduce user tagging friction when uploading images. AI pre-fills candidate attributes, but community consensus and reputation validation act as the true reliability authority."

Why Strong: Shows engineering maturity by acknowledging AI classification limits.

SECTION 11: IF THE JUDGE SAYS "YOUR DATA WILL BE INACCURATE"

Scripted Response: "Data freshness and reliability are indeed our primary operational challenges. That is why our architecture relies on temporal confidence decay, multi-user consensus thresholds, and timestamping rather than assuming static node accuracy."

SECTION 12: IF THE JUDGE SAYS "YOU CANNOT BUILD THIS / TOO COMPLEX"

Scripted Response: "We are not proposing nationwide city deployment on day one. Our MVP targets a constrained, high-impact zone—a single university campus or 2 metro interchange stations—using OpenStreetMap geometry and a focused PostGIS routing engine."

SEC 13

REAL-WORLD CASE STUDY: METRO INTERCHANGE HUB

Consider navigating a complex multi-level Metro Interchange Station (e.g., Rajiv Chowk Metro Hub):

STANDARD ROUTE (Shortest / Flawed)

Entrance Gate 1 → Main Concourse Escalator → Platform Staircase (18 Steps) → **DEAD END FOR WHEELCHAIR**

✖ Result: User stranded at stairs; forced to backtrack.

ACCESSICITY ROUTE (Step-Free)

Gate 3 Accessible Ramp → North Lift B (Verified Operational 10m ago) → Smooth Corridor → Level Platform Access

✔ Result: 100% usable, seamless physical navigation.

Hinglish Summary: Standard route chhota hai (100 meters) par usme sidhi hai. AccessiCity route thoda lamba hai (150 meters) par usme working lift aur ramp hai. User pahunch sake, yeh zaroori hai!

SEC 14

COLOURFUL MEMORY WALL (LINES TO MEMORIZE)

"Google Maps tells you where to go. AccessiCity tells you whether you can actually get there."

"Shortest route aur usable route same cheez nahi hain."

"AI assists the data pipeline; it does not become the final authority."

"Accessibility is a first-class routing constraint."

Avoid making arrogant or technically unviable claims during judge questioning. Stick strictly to verified project positioning:

✗ NEVER SAY (Red Flags)

- "Google Maps has no accessibility data at all." (*Incorrect, Google has basic wheelchair flags*).
- "Our AI is 100% accurate in detecting ramps." (*No vision model is 100% perfect*).
- "We have live hardware sensors on every elevator." (*Unrealistic initial claim*).
- "We will map the entire country immediately." (*Shows lack of MVP focus*).
- "Nobody else can copy this idea." (*Idea is defensible via data network effect, not magic*).

✓ INSTEAD SAY (Judge Safe)

- "AccessiCity focuses specifically on graph-bound accessibility constraints."
- "Computer vision assists data ingestion, while community consensus validates accuracy."
- "We track status via temporal crowdsourced telemetry and partner transit feeds."
- "Our MVP focuses on high-density transit hubs first before scaling."
- "Our moat lies in hyper-local verified data assets and community network effects."

WARNING: Overclaiming features causes judges to aggressively test implementation details. Honest boundaries earn respect.

1. PROBLEM

Inaccessible stairs/broken lifts make standard routes physically unusable.

2. SOLUTION

Accessibility-first navigation mapping step-free paths using crowd verification.

3. TARGET USERS

Wheelchair users, baby stroller parents, and mobility-constrained citizens.

4. BASE MAP DATA

OpenStreetMap (OSM) vector geometry and custom PostGIS tag overlays.

5. AI ROLE

CV Model assists by detecting ramps in user photos to pre-fill reports.

6. VERIFICATION

Multi-user consensus, temporal decay scoring, and contributor reputation.

7. ROUTING MATH

Graph $G=(V, E)$ search algorithm applying ∞ cost penalty to stair edges.

8. FRONTEND

React Native / Mapbox GL app with Step-Free toggle and photo upload sheet.

9. BACKEND API

Node.js Express / Python FastAPI REST endpoints returning GeoJSON routes.

10. DATABASE

PostgreSQL + PostGIS storing spatial ramps, lifts, logs, and reputation.

11. MVP SCOPE

Galgotias University Campus + 2 Metro Interchange hubs.

12. CORE INNOVATION

Integrating crowdsourced micro-accessibility directly into graph routing engines.

FINAL MOTTO TO REMEMBER

"DON'T JUST FIND A ROUTE. FIND A ROUTE YOU CAN ACTUALLY USE."